

# Multimodal Transformer-Based Arousal Estimation from Physiological Signals

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**Abstract**—We present an investigation into binary arousal estimation from peripheral physiological signals using a multimodal cross-attention transformer, evaluated against handcrafted feature baselines. Our architecture, EmotionalTransformer, combines modality-specific multi-scale 1D convolutional encoders with a cross-attention fusion mechanism, and accepts three peripheral signals—photoplethysmography (PPG), galvanic skin response (GSR), and respiration—alongside 55 handcrafted time-domain, frequency-domain, and nonlinear features. We evaluate on the CASE and DEAP datasets under leave-one-subject-out (LOSO) cross-validation. Despite careful design choices including self-supervised pre-training via masked signal reconstruction and class-imbalance correction via weighted loss, a linear SVM trained on handcrafted features outperformed the transformer on peripheral signals, achieving 66.1% accuracy on CASE versus the transformer’s 60.9%. Our results are consistent with recent benchmark findings that classical methods remain competitive or superior to deep learning on peripheral-only emotion recognition. We analyze the role of class imbalance, cross-dataset transfer failures, and inter-subject variance, and discuss paths forward including gradient-boosted classifiers, contrastive self-supervised pre-training, and EEG integration.

**Index Terms**—arousal estimation, emotion recognition, multimodal learning, physiological signals, transformer, cross-attention

## I. INTRODUCTION

Arousal—the dimension of emotional experience corresponding to activation or excitement level—plays a central role in affective computing, human-computer interaction, and mental health monitoring. Unlike valence, which is notoriously difficult to decode from peripheral physiology alone, arousal has a stronger coupling to the autonomic nervous system and is reflected in signals such as galvanic skin response (GSR), heart rate (via PPG), and respiratory patterns [1]. Reliable automated arousal estimation from wearable-grade peripheral sensors could enable real-time emotional feedback applications ranging from adaptive user interfaces to stress monitoring in clinical settings.

Two paradigms dominate the literature: (1) handcrafted feature engineering followed by classical classifiers such as support vector machines (SVMs) or random forests, and (2) end-to-end deep learning models that learn representations

directly from raw physiological time series. In recent years, transformer-based architectures have demonstrated strong performance on sequential physiological data, partly because their self-attention mechanisms can capture long-range temporal dependencies more flexibly than recurrent networks [4]. Multimodal transformers in particular have shown promise for fusing heterogeneous physiological modalities via cross-attention [5].

Despite these advances, peripheral-only emotion recognition remains challenging. A comprehensive 2023 benchmark of 114 method comparisons across peripheral physiological datasets found that handcrafted features outperformed deep learning in 107 of those comparisons [6]. This discrepancy is attributed to small dataset sizes, high inter-subject variability, and the relative informativeness of frequency-domain and entropy-based features for peripheral signals.

In this work, we investigate whether a compact, carefully regularized transformer architecture can bridge this gap on the CASE [3] and DEAP [2] datasets. Our contributions are:

- 1) A multimodal cross-attention transformer (EmotionalTransformer) with modality-specific multi-scale CNN encoders and a handcrafted feature injection module, totaling approximately 196K parameters.
- 2) An extended handcrafted feature set of 55 features incorporating nonlinear complexity measures—approximate entropy (ApEn), sample entropy (SaEn), fuzzy entropy (FuEn), wavelet packet entropy (WpEn), Higuchi fractal dimension (HFD), Katz fractal dimension (KFD), Petrosian fractal dimension, and Poincaré plot descriptors—alongside standard time- and frequency-domain statistics.
- 3) A masked signal reconstruction pre-training strategy and systematic evaluation of class-imbalance correction strategies under LOSO cross-validation on two peripheral physiological datasets.
- 4) An empirical analysis of failure modes including cross-dataset transfer and the impact of feature-count mismatches during pre-training.

## II. RELATED WORK

### A. Physiological Emotion Recognition

Early work on physiological emotion recognition relied heavily on handcrafted features extracted from electrodermal activity (EDA), blood volume pulse (BVP), respiration, and skin temperature, fed into SVMs or linear discriminant analysis [1]. The DEAP dataset [2] established a widely-used benchmark for valence-arousal prediction from peripheral and EEG signals, though a 2025 review found that 57.4% of DEAP-based studies employ evaluation protocols susceptible to data leakage or invalid train-test splits [9]. The CASE dataset [3] provides continuous valence-arousal annotations and is designed for naturalistic emotion elicitation, making it particularly relevant for realistic evaluation scenarios.

Fractal and entropy-based features have demonstrated strong discriminative power for arousal. Mercado-Diaz et al. [7] achieved 84.3% classification accuracy on the CASE dataset using EDA fractal features, underscoring the informativeness of nonlinear complexity measures. More broadly, the FEEL benchmark [6] systematically compared 114 feature-classifier combinations on peripheral datasets and found classical pipelines to be consistently superior to deep models under subject-independent evaluation.

### B. Transformer-Based Emotion Modeling

Transformers have been adapted to physiological time series in several forms. Vanilla transformers applied patch-wise to EEG have achieved competitive performance on large-scale datasets [4]. Architectures such as the rU-Net with self-supervised learning demonstrated 70.3% LOSO accuracy on peripheral signals [8], providing a realistic performance ceiling for current methods on modality-limited settings. Recurrent transformer hybrids [5] and temporal convolutional encoders with self-attention pooling have also been explored. Our work differs in that we deliberately keep the transformer compact (196K parameters) and supplement learned representations with injected handcrafted features, aiming for a hybrid model that captures the strengths of both paradigms.

### C. Attention-Based Multimodal Fusion

Multimodal physiological emotion recognition typically fuses signals at the feature level (concatenation), decision level (ensemble), or representation level (attention-based). Cross-attention has emerged as a principled mechanism for representation-level fusion: one modality’s queries attend over another modality’s keys and values, enabling the model to selectively integrate relevant cross-modal information [5]. Prior work has demonstrated cross-attention fusion for EEG-peripheral combinations and for audio-visual emotion tasks. Our architecture extends this to three peripheral modalities (PPG, GSR, respiration), using a symmetric cross-attention scheme within a compact transformer block.

## III. METHODS

### A. Problem and Data

We formulate arousal estimation as binary classification. Continuous arousal annotations are binarized using a per-subject median split: windows above the subject’s median arousal score are labeled high-arousal (1) and those at or below are labeled low-arousal (0). Per-subject splitting prevents individual baseline differences from inflating inter-class variance and is consistent with standard practice for within-subject normalization in affective computing [2].

**CASE dataset.** The CASE dataset [3] contains continuous valence-arousal annotations from 30 subjects watching video clips, with peripheral physiological signals sampled at 1000 Hz. After windowing at 10 seconds with 5-second stride (evaluation) or 2.5-second stride (pre-training), we obtained 13,950 windows. The resulting class distribution is 76.1% high-arousal and 23.9% low-arousal, reflecting a significant class imbalance due to the stimuli design.

**DEAP dataset.** The DEAP dataset [2] provides EEG and peripheral physiological recordings from 32 subjects; we use only the 23 subjects for whom peripheral data (PPG, GSR, respiration) are available. After the same windowing procedure, we obtained 10,077 windows with a class distribution of 58.7% high-arousal and 41.3% low-arousal.

Both datasets were evaluated under leave-one-subject-out (LOSO) cross-validation to measure subject-independent generalization. For computational tractability, we use a 10-fold LOSO sample (10 held-out subjects drawn from the 30) on CASE and full LOSO on DEAP.

### B. Handcrafted Feature Set

For each 10-second window of each signal, we extract 55 features organized into two groups:

**Time- and frequency-domain features (28).** These include statistical moments (mean, variance, skewness, kurtosis), zero-crossing rate, energy, peak-to-peak amplitude, band power estimates, spectral centroid, spectral entropy, and dominant frequency. These features are standard in physiological signal analysis [1] and provide a compact summary of linear signal dynamics.

**Nonlinear complexity features (27).** We extend the standard feature set with entropy-based and fractal-based complexity measures:

- *Approximate Entropy (ApEn)* and *Sample Entropy (SaEn)*: quantify the regularity and unpredictability of the time series, with SaEn being less biased for short sequences [10].
- *Fuzzy Entropy (FuEn)*: a fuzzy membership function-based variant that is more robust to noise and parameter sensitivity [11].
- *Wavelet Packet Entropy (WpEn)*: measures complexity in the wavelet domain, capturing multi-resolution structure.
- *Higuchi Fractal Dimension (HFD)* and *Katz Fractal Dimension (KFD)*: characterize the self-similarity and complexity of the waveform in the time domain.

- *Petrosian Fractal Dimension*: a computationally efficient fractal estimator based on the number of sign changes in the first-difference series.
- *Poincaré SD1, SD2, and SD1/SD2 ratio*: derived from Poincaré plot analysis of the signal, capturing short-term and long-term variability alongside their ratio.

Each feature is computed per-signal per-window, yielding a concatenated 55-dimensional feature vector (across PPG, GSR, and respiration channels). These features are used both as the input to the SVM baseline and as auxiliary input to the handcrafted feature injection module of the transformer.

### C. CNN Encoders

Each peripheral signal is processed by a modality-specific multi-scale 1D CNN encoder. The encoder applies three parallel convolutional branches with kernel sizes of 7, 31, and 129 to capture short-range, medium-range, and long-range temporal patterns, respectively. Each branch consists of a 1D convolution (with  $C = 16$  output channels), batch normalization, and ReLU activation. The outputs of the three branches are concatenated along the channel dimension and passed through a final projection layer, producing a fixed-length embedding of dimension  $d_{\text{model}} = 32$  for each modality. This multi-scale design is motivated by the heterogeneous temporal scales present in physiological signals: GSR dynamics operate over several seconds, whereas PPG oscillations occur at sub-second intervals.

### D. Multimodal Transformer with Cross-Attention

**Architecture overview.** The EmotionalTransformer fuses the three CNN embeddings (from PPG, GSR, and respiration) with the handcrafted feature vector via a cross-attention transformer. The architecture operates in two stages:

- 1) **Handcrafted feature injection (hc\_injector)**: The 55-dimensional handcrafted feature vector is projected via a learned linear layer to  $d_{\text{model}} = 32$ , producing a single feature token that is concatenated with the three CNN embedding tokens.
- 2) **Cross-attention transformer block**: A standard transformer encoder layer with  $H = 4$  attention heads,  $d_{\text{model}} = 32$ , and a feed-forward dimension of 128 operates over the four-token sequence (three CNN tokens plus one handcrafted feature token). This allows each modality to attend to all others, enabling the model to learn context-dependent fusion weights.

The transformer block uses pre-layer normalization and dropout (rate 0.1). The output token sequence is mean-pooled and passed through a two-layer classification head with a hidden dimension of 32, producing a single logit for binary classification. With the “small” preset ( $d_{\text{model}} = 32$ ,  $C_{\text{CNN}} = 16$ , 4 heads, 1 transformer layer), the total parameter count is approximately 196K. This deliberately compact design limits overfitting on the limited training data available in LOSO folds.

**Self-supervised pre-training.** Prior to supervised fine-tuning, the encoder and transformer are pre-trained on the

CASE dataset using a masked signal reconstruction objective. A randomly selected 15% of time steps in each input window are masked (replaced with zeros), and the model is trained to reconstruct the original signal values at masked positions via mean squared error (MSE) loss. This encourages the encoder to learn robust, generalizable representations before exposure to potentially noisy supervision. The pre-training uses AdamW with a learning rate of  $1 \times 10^{-3}$ , weight decay  $1 \times 10^{-4}$ , and a cosine annealing schedule, with a stride of 2.5 seconds for denser sampling.

**Fine-tuning.** Supervised fine-tuning uses binary cross-entropy with logits (BCEWithLogitsLoss). To address the significant class imbalance in CASE (76.1% high-arousal), we apply a positive class weight  $w_{\text{pos}} = N_{\text{neg}}/N_{\text{pos}} \approx 3.18$ , scaling the loss for minority-class samples accordingly. Fine-tuning uses AdamW with a learning rate of  $5 \times 10^{-4}$  and weight decay  $1 \times 10^{-4}$ .

**SVM baseline.** We train a linear SVM on the 55-dimensional handcrafted feature vector, with the regularization parameter  $C$  tuned via inner cross-validation. This baseline serves as the primary reference point for assessing the value-add of the transformer architecture.

## IV. EXPERIMENTS

We conduct a series of experiments to evaluate the impact of pre-training, class-imbalance correction, and cross-dataset transfer. All experiments use LOSO cross-validation, reporting mean  $\pm$  standard deviation across folds. The primary evaluation metrics are classification accuracy, balanced accuracy (average per-class recall), and macro-averaged F1 score. Balanced accuracy and macro F1 are reported for transformer experiments to account for class imbalance effects.

Table I summarizes all experimental results. We compare the SVM baseline to several configurations of the EmotionalTransformer on both CASE and DEAP.

**Experiment 1: SVM baseline.** We trained a linear SVM on the full 55-feature set under LOSO. On CASE, the SVM achieved 66.1% accuracy ( $\pm 15.9\%$ ). An ablation restricted to the original 28 time- and frequency-domain features (not shown in table) achieved approximately 61%, confirming that the nonlinear complexity features (ApEn, SaEn, FuEn, WpEn, HFD) contributed a meaningful gain of  $\sim 5$  percentage points. On DEAP, the SVM achieved 63.1% ( $\pm 14.2\%$ ), indicating that the peripheral features generalize reasonably to the second dataset without any cross-dataset pre-training.

**Experiment 2: Transformer with pre-training and pos\_weight (CASE).** The transformer pre-trained on CASE with masked signal reconstruction and fine-tuned with pos\_weight correction achieved 60.9% accuracy and 65.7% balanced accuracy. The gap between raw accuracy (60.9%) and balanced accuracy (65.7%) reflects the class imbalance: without pos\_weight, the model collapsed to predicting the majority class, yielding  $\sim 76\%$  accuracy but  $\sim 50\%$  balanced accuracy. The pos\_weight correction successfully restored sensitivity to the minority class.

TABLE I  
 LOSO CROSS-VALIDATION RESULTS FOR AROUSAL CLASSIFICATION. BALANCED ACCURACY AND MACRO F1 ARE NOT REPORTED FOR SVM BASELINES WHERE ONLY ACCURACY WAS TRACKED. HIGHER IS BETTER FOR ALL METRICS.

| Model / Configuration                        | Dataset | Accuracy (%)    | Balanced Acc. (%) | Macro F1 (%)    |
|----------------------------------------------|---------|-----------------|-------------------|-----------------|
| SVM (55 features)                            | CASE    | 66.1 $\pm$ 15.9 | —                 | —               |
| SVM (55 features)                            | DEAP    | 63.1 $\pm$ 14.2 | —                 | —               |
| SVM (55 features, LOTO per-subject)          | CASE    | 65.5 $\pm$ 16.0 | —                 | —               |
| Transformer + CASE pretrain + pos_weight     | CASE    | 60.9 $\pm$ 16.8 | 65.7 $\pm$ 15.9   | 55.8 $\pm$ 19.3 |
| Transformer + CASE pretrain + median thresh. | CASE    | 60.9 $\pm$ 13.5 | 65.1 $\pm$ 11.6   | 55.9 $\pm$ 14.7 |
| Transformer + corrected pretrain             | CASE    | 56.4 $\pm$ 16.2 | 62.3 $\pm$ 15.5   | 53.3 $\pm$ 17.6 |
| Transformer + pretrain + pos_weight          | DEAP    | 50.5 $\pm$ 7.5  | 51.2 $\pm$ 7.0    | 46.9 $\pm$ 8.0  |

**Experiment 3: Median thresholding variant.** Replacing the per-subject median split with a fixed global threshold yielded nearly identical accuracy (60.9%) and slightly lower balanced accuracy (65.1%), confirming that the per-subject split is the more robust binarization strategy.

**Experiment 4: Corrected pre-training.** In an earlier experiment, a feature-count mismatch between pre-training (58 features, inadvertently including 3 temperature channels absent in the fine-tuning pipeline) and fine-tuning (55 features) necessitated random re-initialization of the `hc_injector` projection layer. After correcting the feature count to 55 throughout, the pre-trained weights transferred cleanly, but accuracy dropped to 56.4% with a balanced accuracy of 62.3%. This counterintuitive result—where the mismatch configuration outperformed the corrected one—is discussed in the Results and Discussion section.

**Experiment 5: Cross-dataset transfer (DEAP).** Fine-tuning the CASE-pretrained transformer on DEAP yielded only 50.5% accuracy and 51.2% balanced accuracy, barely above chance. The Macro F1 of 46.9% further indicates that the model failed to learn a useful decision boundary under LOSO for DEAP. This result highlights a fundamental cross-dataset transfer challenge.

## V. RESULTS AND DISCUSSION

**SVM versus transformer.** The central finding of this study is that the linear SVM trained on 55 handcrafted features outperformed the transformer in raw accuracy on both CASE (66.1% vs. 60.9%) and in subject-independent robustness. This result is consistent with the FEEL benchmark [6], which found that handcrafted feature pipelines outperformed deep learning methods in 107 of 114 comparisons on peripheral physiological signals. The gap is likely attributable to two factors: the relatively small number of training subjects per LOSO fold (29 subjects for CASE, as few as 22 for DEAP), and the high inter-subject variability that limits the generalization of learned temporal representations.

**Nonlinear features improve SVM.** The addition of nonlinear complexity features (ApEn, SaEn, FuEn, WpEn, HFD) to the 28-feature baseline improved SVM accuracy from approximately 61% to 66.1% on CASE—a meaningful gain. This is consistent with findings by Mercado-Diaz et al. [7], who reported state-of-the-art performance on CASE using EDA fractal features (84.3% with EDA alone, though they used

more subjects and different evaluation conditions). Entropy and fractal features capture signal complexity and irregularity that linear spectral features miss, and these properties are physiologically meaningful: emotional arousal induces changes in the autonomic nervous system that manifest as altered complexity in peripheral signals.

**Class imbalance as a critical factor.** The CASE dataset’s 76.1%/23.9% class split posed a significant challenge for the transformer. Without the `pos_weight` correction, the model collapsed to predicting the majority class in most LOSO folds, achieving near-chance balanced accuracy ( $\sim 50\%$ ) despite superficially high raw accuracy ( $\sim 76\%$ ). The `pos_weight` mechanism (setting  $w_{\text{pos}} \approx 3.18$ ) effectively restored minority-class sensitivity, improving balanced accuracy from  $\sim 50\%$  to 65.7%. This experience highlights the importance of reporting balanced accuracy or macro F1 in imbalanced settings, not just raw accuracy.

**Cross-dataset transfer failure.** The transformer pre-trained on CASE failed to generalize to DEAP, with DEAP balanced accuracy at 51.2%—essentially chance. Several factors likely contribute: (1) the CASE and DEAP datasets use different stimuli and annotation protocols (continuous vs. trial-level, different affective scales), making their arousal distributions incomparable; (2) the sensor hardware and sampling rates differ, altering the morphology of raw signals; (3) the per-subject median split produces different decision boundaries across datasets, so a CASE-pretrained representation may encode dataset-specific spectral artifacts rather than transferable arousal signatures. The SVM, operating on normalized handcrafted features, was more resilient to these distributional shifts (63.1% on DEAP), suggesting that invariant feature representations are more portable than learned raw-signal embeddings.

**Feature-count mismatch and implicit regularization.** The corrected pre-training experiment (Experiment 4) produced a counterintuitive result: correcting the 58-to-55 feature mismatch actually decreased performance relative to the mismatched configuration. We hypothesize that the random re-initialization of the `hc_injector` in the mismatched setup functioned as a form of implicit regularization: the randomly initialized projection was effectively a bottleneck that prevented the model from overfitting the noisy handcrafted features during fine-tuning. When the pre-trained `hc_injector`

weights were transferred (corrected setup), the model may have retained pre-training-specific feature biases that were not aligned with fine-tuning supervision. This finding warrants further investigation and reinforces the importance of careful feature pipeline hygiene.

**High inter-subject variance.** Standard deviations of 13–16% across LOSO folds indicate substantial individual differences in the physiological expression of arousal. Some subjects show high within-session arousal variation well captured by the model (fold accuracy >80%), while others show weak physiological responses that cannot be reliably decoded regardless of model choice. This heterogeneity limits the usefulness of aggregate accuracy as a performance metric and motivates subject-adaptive approaches. Realistic ceiling performance for peripheral LOSO arousal estimation is around 70%, as demonstrated by Chen et al. [8] with the rU-Net using self-supervised learning.

**Where does the transformer add value?** Despite underperforming the SVM in accuracy, the transformer’s balanced accuracy (65.7%) is comparable to, and in minority-class recall arguably better than, the SVM’s raw accuracy (66.1%—which does not account for the imbalanced distribution). The transformer also provides a modular framework for integrating additional modalities (e.g., EEG) via the existing cross-attention mechanism, which the flat SVM feature vector cannot accommodate as naturally. The hybrid handcrafted-feature injection design confirms that explicit domain knowledge remains valuable even within a deep learning framework.

## VI. CONCLUSION

We developed and evaluated EmotionalTransformer, a multimodal cross-attention transformer for binary arousal estimation from peripheral physiological signals (PPG, GSR, respiration). Despite architectural innovations including multi-scale CNN encoders, handcrafted feature injection, masked self-supervised pre-training, and class-imbalance correction, the transformer was outperformed by a linear SVM trained on 55 handcrafted features under leave-one-subject-out cross-validation on the CASE dataset (66.1% vs. 60.9% accuracy). Our results corroborate the FEEL benchmark [6] conclusion that classical feature-based pipelines remain competitive with deep learning on peripheral physiological signals, particularly in low-data, high-variance settings.

Key lessons learned include: (1) class imbalance correction via `pos_weight` is essential for transformer training on imbalanced physiological datasets; (2) nonlinear entropy and fractal features (ApEn, SaEn, FuEn, WpEn, HFD) provide a meaningful boost to SVM performance on peripheral signals; (3) cross-dataset transfer of pre-trained representations is unreliable when annotation protocols and hardware differ; and (4) inter-subject variability remains the dominant bottleneck for subject-independent arousal estimation from peripheral-only signals.

## VII. FUTURE WORK

Several directions hold promise for improving peripheral-only arousal estimation:

**Gradient boosting classifiers.** The most immediate next step is to replace the linear SVM with gradient-boosted classifiers—specifically XGBoost [12] and LightGBM [13]—operating on the 55 handcrafted features. Recent work has demonstrated that XGBoost achieves 88% arousal classification accuracy on DEAP EEG features versus only 60% for SVM under comparable conditions [14], and LightGBM has achieved macro F1 scores of 0.814–0.860 on all 8 peripheral DEAP signals [15]. Combining these classifiers with Bayesian hyperparameter optimization (via Optuna or Hyperopt) could substantially improve on the current SVM baseline without requiring architectural changes.

**EEG integration.** The EmotionalTransformer architecture is already designed to accommodate additional input modalities. Adding EEG—the most informative modality for arousal, with LOSO accuracies regularly exceeding 80% on DEAP [2]—would likely yield the single largest performance gain. The existing cross-attention framework would allow the model to learn which EEG frequencies and spatial channels are most relevant given concurrent peripheral signals, potentially outperforming EEG-only or peripheral-only models through complementary fusion.

**Improved self-supervised pre-training.** The masked signal reconstruction objective used in this work is a relatively weak pre-training signal. Contrastive self-supervised learning—for instance, a SimCLR-style [16] framework in which augmented views of the same window are pulled together in representation space while views from different windows are pushed apart—could learn more discriminative representations prior to fine-tuning. Augmentation strategies appropriate for physiological signals (amplitude scaling, temporal jitter, channel dropout) would need to be designed carefully to avoid collapsing physiologically meaningful variation.

**Subject-adaptive methods.** The high inter-subject variance ( $\sigma \approx 15\%$ ) motivates subject-adaptive inference strategies. Few-shot fine-tuning at test time (using a small calibration set from the held-out subject) or personalized domain adaptation [17] could reduce the effective variance by leveraging a few labeled examples from each new subject, bridging the gap between subject-independent and subject-dependent models.

**Feature selection and dimensionality reduction.** Not all 55 features are equally informative for every subject or signal. Recursive feature elimination (RFE) with cross-validation, mutual information-based selection, or Shapley value analysis (SHAP) could identify the minimal set of features that achieves near-optimal performance, reducing noise and potentially improving generalization. Feature selection is also valuable for interpretability: understanding which entropy or fractal features drive predictions provides insight into the physiological mechanisms underlying arousal differences.

**Larger pre-training corpora.** Pre-training on a single dataset (CASE) limits the diversity of physiological patterns seen during pre-training. Aggregating multiple publicly

available physiological datasets (e.g., MAHNOB-HCI, AMI-GOS, DREAMER) for joint pre-training, followed by dataset-specific fine-tuning, could yield more robust pretrained representations and improve cross-dataset transfer—addressing the near-chance DEAP performance observed in this work.

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